

Week 6 recitation

Fishery (a): sampling

Fifteen observations were generated from each distribution. The discrete probabilities are 0.20, 0.10, 0.50, 0.10, and 0.10 for outcomes 0, 1, 2, 3, and 4. Independent uniform streams are stored in columns N:S and transformed with Excel formulas.

U(0,1)	Discrete	Bin(15,0.7)	U(5,9)	N(5,2)	Exp(0.4)
0.7311	2	8	6.2428	3.9296	1.7510
0.8357	0	13	8.6414	2.6228	1.7157
0.0275	2	11	7.7720	2.8757	1.8709
0.0704	0	11	7.7356	7.9012	3.3823
0.1045	0	12	5.4996	5.3155	1.6168
0.4317	2	10	7.1175	5.4499	8.1033
0.0486	0	7	6.2982	4.2319	0.6278
0.7888	2	10	8.9414	3.6067	1.9448
0.2247	2	10	6.6190	7.5360	0.1689
0.2537	2	9	8.0868	6.7551	2.1376
0.2683	0	10	7.0744	4.6115	1.3474
0.6082	4	11	5.7140	6.2452	7.9281
0.1148	1	12	6.7303	6.6309	0.6419
0.1398	0	11	6.5819	4.9439	3.4729
0.9024	4	12	8.2108	3.3690	1.4469

Excel capture: Sampling!G2:L17. All 15 observations are shown. N(5,2) uses standard deviation 2; Exp(0.4) uses rate 0.4, with mean 2.5.

Distribution	Excel formula, first observation
Uniform (0,1)	=N3
Discrete	=LOOKUP(O3,\$E\$3:\$E\$7,\$B\$3:\$B\$7)
Binomial (15,0.7)	=BINOM.INV(15,0.7,P3)
Uniform (5,9)	=5+4*Q3
Normal (5,2)	=NORM.INV(R3,5,2)
Exponential (rate 0.4)	=-LN(1-S3)/0.4

Uniform draws were generated with NumPy seed 76006 and frozen as workbook inputs. Transformations, profits, summaries, and confidence limits are Excel formulas. Captures come from the completed Excel workbook; Solver images show the actual dialogs.

Reference: RS W6.pdf, RS W6 Sol.pdf and RS W6 Sol.xlsx were used to check the intended model and workbook structure. Simulation draws and statistics are our own realization.

Fishery (b): 70-day simulation

Daily price is Normal(\$3.65, \$0.20). Demand is sampled independently of price. Sold = MIN(3500,Demand);
Revenue = Price x Sold; Profit = Revenue - \$10,000.

Days	Mean profit	Sample SD	95% CI for mean
70	\$2,084.20	\$1,837.20	\$1,646.14 to \$2,522.27

Mean = AVERAGE(O4:O73), SD = STDEV.S(O4:O73), and CI = mean +/- T.INV.2T(0.05,69) x SD / SQRT(70). The interval estimates expected daily profit.

Day	Price RND	Price	Demand RND	Demand	Sold	Revenues	Cost	Profit
1	0.7311	3.7732	0.6506	5000	3500	\$13,206.34	\$10,000.00	\$3,206.34
2	0.1253	3.4202	0.3107	4000	3500	\$11,970.75	\$10,000.00	\$1,970.75
3	0.2963	3.5430	0.5036	4000	3500	\$12,400.36	\$10,000.00	\$2,400.36
4	0.8357	3.8454	0.1287	3000	3000	\$11,536.20	\$10,000.00	\$1,536.20
5	0.8947	3.9004	0.9103	6000	3500	\$13,651.35	\$10,000.00	\$3,651.35
6	0.1173	3.4123	0.4966	4000	3500	\$11,942.97	\$10,000.00	\$1,942.97
7	0.0275	3.2661	0.7962	5000	3500	\$11,431.41	\$10,000.00	\$1,431.41
8	0.6635	3.7344	0.6930	5000	3500	\$13,070.37	\$10,000.00	\$3,070.37
9	0.1441	3.4376	0.5269	5000	3500	\$12,031.51	\$10,000.00	\$2,031.51
10	0.0704	3.3554	0.0549	2000	2000	\$6,710.74	\$10,000.00	(\$3,289.26)
11	0.6759	3.7413	0.6839	5000	3500	\$13,094.46	\$10,000.00	\$3,094.46
12	0.9266	3.9401	0.7415	5000	3500	\$13,790.42	\$10,000.00	\$3,790.42
13	0.1045	3.3987	0.0465	1000	1000	\$3,398.75	\$10,000.00	(\$6,601.25)
14	0.7446	3.7815	0.1249	3000	3000	\$11,344.49	\$10,000.00	\$1,344.49
15	0.5627	3.6815	0.4762	4000	3500	\$12,885.41	\$10,000.00	\$2,885.41
16	0.4317	3.6156	0.5053	4000	3500	\$12,654.65	\$10,000.00	\$2,654.65
17	0.3525	3.5743	0.5294	5000	3500	\$12,509.92	\$10,000.00	\$2,509.92
18	0.5890	3.6950	0.9609	6000	3500	\$12,932.45	\$10,000.00	\$2,932.45
19	0.0486	3.3182	0.0764	2000	2000	\$6,636.40	\$10,000.00	(\$3,363.60)
20	0.0231	3.2514	0.3246	4000	3500	\$11,380.02	\$10,000.00	\$1,380.02
21	0.3505	3.5732	0.2221	4000	3500	\$12,506.18	\$10,000.00	\$2,506.18
22	0.7888	3.8105	0.5604	5000	3500	\$13,336.63	\$10,000.00	\$3,336.63
23	0.3127	3.5524	0.9853	6000	3500	\$12,433.25	\$10,000.00	\$2,433.25
24	0.2430	3.5107	0.5406	5000	3500	\$12,287.36	\$10,000.00	\$2,287.36
25	0.2247	3.4987	0.5298	5000	3500	\$12,245.46	\$10,000.00	\$2,245.46
26	0.4145	3.6068	0.4047	4000	3500	\$12,623.82	\$10,000.00	\$2,623.82
27	0.8976	3.9036	0.0653	2000	2000	\$7,807.20	\$10,000.00	(\$2,192.80)
28	0.2537	3.5174	0.3092	4000	3500	\$12,310.91	\$10,000.00	\$2,310.91
29	0.2164	3.4931	0.7717	5000	3500	\$12,225.85	\$10,000.00	\$2,225.85
30	0.8099	3.8255	0.5747	5000	3500	\$13,389.28	\$10,000.00	\$3,389.28
31	0.2683	3.5264	0.1877	4000	3500	\$12,342.44	\$10,000.00	\$2,342.44
32	0.3503	3.5731	0.5186	5000	3500	\$12,505.89	\$10,000.00	\$2,505.89
33	0.4230	3.6112	0.4167	4000	3500	\$12,639.03	\$10,000.00	\$2,639.03
34	0.6082	3.7049	0.9703	6000	3500	\$12,967.26	\$10,000.00	\$2,967.26
35	0.6362	3.7197	0.1785	3000	3000	\$11,159.02	\$10,000.00	\$1,159.02

Excel capture: Fishery operations!G3:O38. Days 1-35; the remaining 35 days appear on the next page.

Fishery (b): days 36-70

36	0.7332	3.7745	0.9580	6000	3500	\$13,210.82	\$10,000.00	\$3,210.82
37	0.1148	3.4098	0.2906	4000	3500	\$11,934.19	\$10,000.00	\$1,934.19
38	0.7470	3.7830	0.4326	4000	3500	\$13,240.52	\$10,000.00	\$3,240.52
39	0.7926	3.8131	0.2264	4000	3500	\$13,345.82	\$10,000.00	\$3,345.82
40	0.1398	3.4338	0.1451	3000	3000	\$10,301.37	\$10,000.00	\$301.37
41	0.5881	3.6945	0.3955	4000	3500	\$12,930.80	\$10,000.00	\$2,930.80
42	0.4888	3.6444	0.7507	5000	3500	\$12,755.37	\$10,000.00	\$2,755.37
43	0.9024	3.9090	0.9212	6000	3500	\$13,681.55	\$10,000.00	\$3,681.55
44	0.8497	3.8571	0.8027	6000	3500	\$13,499.68	\$10,000.00	\$3,499.68
45	0.2074	3.4869	0.4394	4000	3500	\$12,204.15	\$10,000.00	\$2,204.15
46	0.0787	3.3673	0.1725	3000	3000	\$10,101.79	\$10,000.00	\$101.79
47	0.8991	3.9053	0.5691	5000	3500	\$13,668.63	\$10,000.00	\$3,668.63
48	0.1184	3.4133	0.5420	5000	3500	\$11,946.71	\$10,000.00	\$1,946.71
49	0.1115	3.4063	0.7931	5000	3500	\$11,922.05	\$10,000.00	\$1,922.05
50	0.3498	3.5728	0.6664	5000	3500	\$12,504.83	\$10,000.00	\$2,504.83
51	0.2300	3.5023	0.4376	4000	3500	\$12,257.90	\$10,000.00	\$2,257.90
52	0.2703	3.5276	0.3942	4000	3500	\$12,346.70	\$10,000.00	\$2,346.70
53	0.3712	3.5843	0.0849	2000	2000	\$7,168.50	\$10,000.00	(\$2,831.50)
54	0.9113	3.9198	0.7106	5000	3500	\$13,719.30	\$10,000.00	\$3,719.30
55	0.5854	3.6931	0.4822	4000	3500	\$12,925.99	\$10,000.00	\$2,925.99
56	0.2261	3.4996	0.9015	6000	3500	\$12,248.72	\$10,000.00	\$2,248.72
57	0.8355	3.8452	0.4952	4000	3500	\$13,458.23	\$10,000.00	\$3,458.23
58	0.3852	3.5917	0.4219	4000	3500	\$12,570.78	\$10,000.00	\$2,570.78
59	0.6295	3.7161	0.4844	4000	3500	\$13,006.33	\$10,000.00	\$3,006.33
60	0.6477	3.7258	0.6038	5000	3500	\$13,040.32	\$10,000.00	\$3,040.32
61	0.4609	3.6303	0.1396	3000	3000	\$10,891.03	\$10,000.00	\$891.03
62	0.5488	3.6745	0.8570	6000	3500	\$12,860.81	\$10,000.00	\$2,860.81
63	0.4144	3.6067	0.5433	5000	3500	\$12,623.56	\$10,000.00	\$2,623.56
64	0.7294	3.7722	0.4898	4000	3500	\$13,202.65	\$10,000.00	\$3,202.65
65	0.1698	3.4590	0.5178	5000	3500	\$12,106.42	\$10,000.00	\$2,106.42
66	0.6558	3.7302	0.3892	4000	3500	\$13,055.81	\$10,000.00	\$3,055.81
67	0.0712	3.3567	0.2093	4000	3500	\$11,748.41	\$10,000.00	\$1,748.41
68	0.6573	3.7310	0.7491	5000	3500	\$13,058.62	\$10,000.00	\$3,058.62
69	0.9646	4.0114	0.5307	5000	3500	\$14,039.98	\$10,000.00	\$4,039.98
70	0.5048	3.6524	0.1761	3000	3000	\$10,957.22	\$10,000.00	\$957.22

Excel capture: Fishery operations!G39:O73. Columns match the preceding page: Day, Price RND, Price, Demand RND, Demand, Sold, Revenues, Cost, Profit.

Demand distribution				Mean	
x	Prob.	CDF	Lower	\$2,084.20	
0	0.02	0.02	0	Standard deviation	
1,000	0.03	0.05	0.02	\$1,837.20	
2,000	0.05	0.1	0.05	CI	
3,000	0.08	0.18	0.1	Lower 95% CI	
4,000	0.33	0.51	0.18	\$1,646.14	
5,000	0.29	0.8	0.51	Upper 95% CI	
6,000	0.2	1	0.8	\$2,522.27	

Fishery operations!B2:E10.

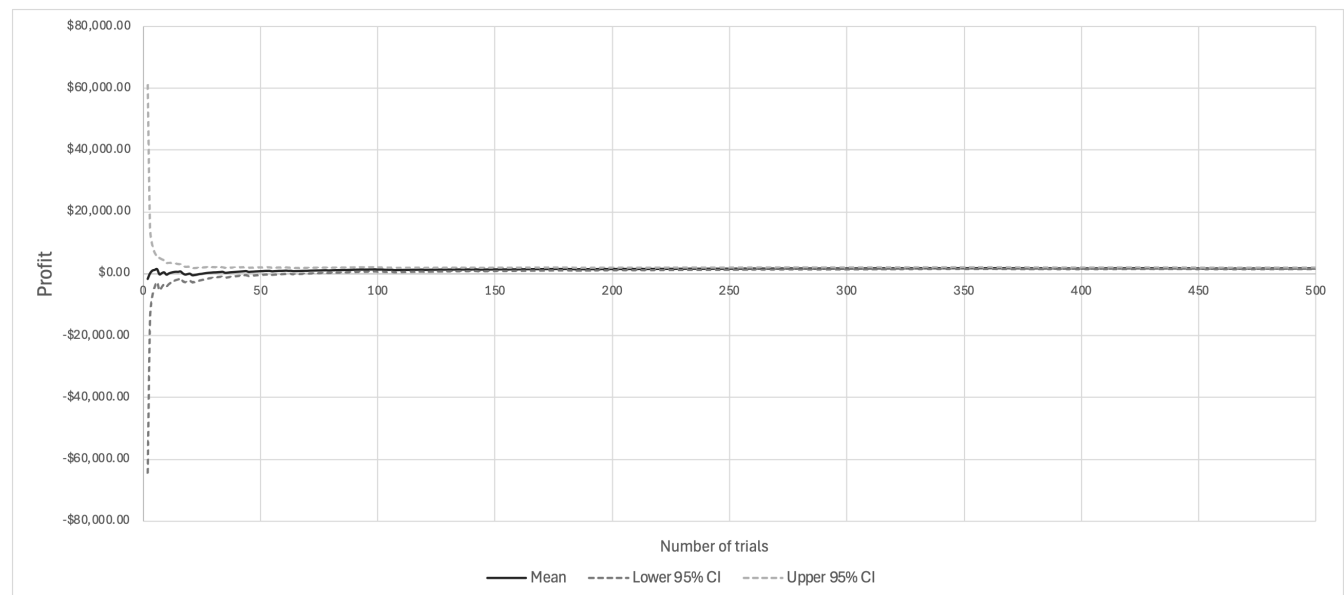
Fishery operations!X4:Y9.

Exact validation: $E[\text{sold}] = 0.02(0) + 0.03(1000) + 0.05(2000) + 0.08(3000) + 0.82(3500) = 3,240$ fish. $E[\text{profit}] = 3.65 \times 3,240 - 10,000 = \$1,826.00$, inside the simulation confidence interval.

Fishery (c): increasing the sample size

Five hundred days were simulated. Each row uses day 1 through its own day n. Mean = AVERAGE(\$O\$4:Or), SD = STDEV.S(\$O\$4:Or), and CI = mean +/- T.INV.2T(0.05,n-1) x SD / SQRT(n). At n = 1, SD and CI are undefined and left blank.

Days	Mean profit	Sample SD	95% CI for mean
500	\$1,822.20	\$2,762.56	\$1,579.46 to \$2,064.93



Native Excel chart from N analysis. The very wide initial interval reflects the small sample size. All 500 days and cumulative calculations are included in the workbook.

Final cumulative results

491	\$1,809.89	\$2,785.09	\$1,562.94	\$2,056.85
492	\$1,810.94	\$2,782.35	\$1,564.48	\$2,057.40
493	\$1,811.65	\$2,779.56	\$1,565.68	\$2,057.61
494	\$1,813.65	\$2,777.10	\$1,568.16	\$2,059.15
495	\$1,816.56	\$2,775.04	\$1,571.50	\$2,061.63
496	\$1,816.84	\$2,772.25	\$1,572.27	\$2,061.41
497	\$1,817.63	\$2,769.51	\$1,573.55	\$2,061.71
498	\$1,821.44	\$2,768.02	\$1,577.74	\$2,065.14
499	\$1,822.43	\$2,765.33	\$1,579.20	\$2,065.65
500	\$1,822.20	\$2,762.56	\$1,579.46	\$2,064.93

N analysis!P494:T503, days 491-500. Columns: n, Mean, Sample SD, Lower 95% CI, Upper 95% CI.

The running mean need not converge monotonically, but its uncertainty decreases as n grows. The final mean of \$1,822.20 is close to the exact expectation of \$1,826.00. The intervals describe the mean, not daily-profit percentiles.

Fishery (d): choosing the quantity Q

Use the quantity-dependent cost in the solution workbook: $\text{Cost}(Q) = \$3,000 + \$2Q$. Profit = Price x MIN(Q,Demand) - Cost(Q). A native one-input Excel Data Table evaluates Q = 2,000 to 6,000 in steps of 100, using the same 400 draws for every Q.

Fishing	\$
2000	-
2100	2,205
2200	-\$33.78
2300	\$92.04
2400	\$217.86
2500	\$343.68
2600	\$469.50
2700	\$595.32
2800	\$721.14
2900	\$846.96
3000	\$972.77
3100	\$1,098.59
3200	\$1,224.41
3300	\$1,322.45
3400	\$1,420.50
3500	\$1,518.54
3600	\$1,616.58
3700	\$1,714.62
3800	\$1,812.66
3900	\$1,910.70
4000	\$2,008.75
4100	\$2,106.79
4200	\$2,204.83
4300	\$2,186.01
4400	\$2,167.20
4500	\$2,148.38
4600	\$2,129.57
4700	\$2,110.75
4800	\$2,091.94
4900	\$2,073.12
5000	\$2,054.31
5100	\$2,035.49
5200	\$2,016.68
5300	\$1,902.43
5400	\$1,788.19
5500	\$1,673.94
5600	\$1,559.70
5700	\$1,445.45
5800	\$1,331.21
5900	\$1,216.96
6000	\$1,102.72
	\$988.47
	\$874.23

Inputs and summary

INPUTS

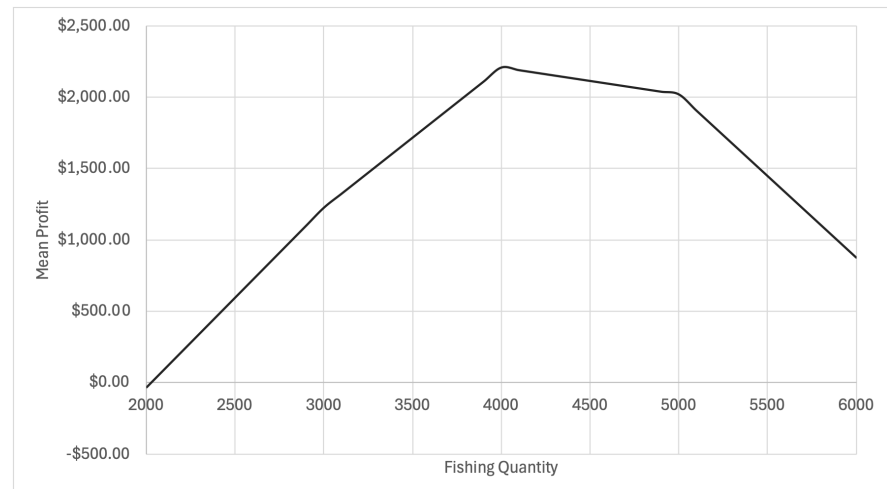
Q (fishes) 4000

OUTPUTS

Mean profit \$2,204.83
 Standard dev \$3,343.04
 Lower 95% CI \$1,876.22
 Upper 95% CI \$2,533.44
 n 400

Fixed cost 3000
 Cost per fish 2

Profit as a function of Q



Recommendation: Q = 4,000 fish. Simulated mean profit is \$2,204.83. Exact expected profit is $3.65 \times 3,650 - (3,000 + 2 \times 4,000) = \$2,322.50$.

Beyond 4,000, marginal expected revenue falls below the \$2 cost per additional fish.

T3:U44: native Data Table.

Excel captures: Analysis on Q!Q3:R14, T3:U44, and its chart. Output U3 = R7; input column T4:T44; column input cell R4. R7 averages O4:O403, so n = 400.